

Cell Biology & Metabolic Biochemistry: Cellular Energetics & Respiration

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1. Structural Organization of the Eukaryotic Cell

The cell represents the fundamental structural and functional unit of all living organisms. Eukaryotic cells are distinguished by membrane-bound organelles that partition biochemical processes into distinct micro-environments. The nucleus houses genomic DNA packaged into chromatin by histone proteins. The Endoplasmic Reticulum (ER) includes the Rough ER, studded with ribosomes synthesizing membrane and secretory proteins, and the Smooth ER, which synthesizes lipids, phospholipids, and steroid hormones while detoxifying metabolic byproducts. The Golgi apparatus modifies, sorts, and packages glycoproteins into transport vesicles targeted for cellular secretion or lysosomes.

2. Thermodynamics and Enzyme Kinetics

Biological chemical reactions obey the laws of thermodynamics. Gibbs free energy change (ΔG) dictates reaction spontaneity: exergonic reactions release free energy ($\Delta G < 0$), whereas endergonic reactions require energy input ($\Delta G > 0$). Enzymes are biological catalysts that increase reaction velocity by lowering the activation energy barrier without altering the reaction equilibrium or net ΔG . The Michaelis-Menten kinetic model describes enzyme reaction rates: $V_0 = (V_{\max} * [S]) / (K_m + [S])$, where $V_{\max}$ is the maximum reaction velocity, $[S]$ is substrate concentration, and K_m (Michaelis constant) represents the substrate concentration at which reaction rate is half of $V_{\max}$. A lower K_m reflects higher enzyme-substrate affinity.

3. Cellular Respiration Stage 1: Glycolysis

Cellular respiration is the catabolic pathway that oxidizes organic fuels to generate adenosine triphosphate (ATP). Glycolysis occurs in the cytoplasm and does not require molecular oxygen. It converts one molecule of 6-carbon glucose into two molecules of 3-carbon pyruvate through 10 enzymatic reactions. The pathway consists of two phases: 1) Energy Investment Phase: Two ATP molecules are consumed by hexokinase and phosphofructokinase-1 (PFK-1, the key rate-limiting committed step). 2) Energy Payoff Phase: Four ATP molecules are generated via substrate-level phosphorylation, alongside two molecules of NADH. Thus, the net yield of glycolysis is 2 ATP, 2 NADH, and 2 pyruvate molecules per glucose.

4. The Citric Acid Cycle (Krebs Cycle)

In the presence of oxygen, pyruvate translocates into the mitochondrial matrix. The pyruvate dehydrogenase complex executes oxidative decarboxylation, converting pyruvate into 2-carbon Acetyl-CoA, producing 1 CO₂ and 1 NADH per pyruvate. Acetyl-CoA enters the Citric Acid Cycle by condensing with 4-carbon oxaloacetate to form 6-carbon citrate, catalyzed by citrate synthase. Across one complete turn of the cycle, two carbons are released as CO₂, yielding 3 NADH, 1 FADH₂, and 1 ATP (or GTP) via substrate-level phosphorylation. Because one glucose yields two Acetyl-CoA molecules, the total Krebs cycle yield per glucose is 6 NADH, 2 FADH₂, 2 ATP, and 4 CO₂.

5. Oxidative Phosphorylation and ATP Synthase

Oxidative phosphorylation occurs at the inner mitochondrial membrane and produces the vast majority of cellular ATP. The Electron Transport Chain (ETC) comprises four multiprotein complexes (Complex I, II, III, and IV). High-energy electrons donated by NADH (at Complex I) and FADH₂ (at Complex II) pass through coenzyme Q (ubiquinone) and cytochrome c to Complex IV, where molecular oxygen serves as the terminal electron acceptor, reducing to form water (H₂O).

As electrons flow exergonically through Complexes I, III, and IV, protons (H⁺) are actively pumped from the mitochondrial matrix into the intermembrane space, generating a steep electrochemical proton gradient (proton-motive force). Peter Mitchell's chemiosmotic hypothesis explains how ATP synthase harnesses this potential energy. Protons flow down their electrochemical gradient through the F₀ channel of ATP synthase, driving rotational conformational changes in the F₁ catalytic subunit that phosphorylate ADP and inorganic phosphate (Pi) into ATP. Theoretical maximum ATP yield from complete aerobic respiration is 30 to 32 ATP molecules per oxidized glucose molecule.

6. Fermentation: Anaerobic ATP Generation

When molecular oxygen is absent, oxidative phosphorylation ceases and NADH cannot transfer electrons to the ETC. To sustain glycolysis, cells must regenerate NAD⁺ from NADH through fermentation. In lactic acid fermentation (e.g. human skeletal muscle during intense exercise), lactate dehydrogenase reduces pyruvate to lactate while oxidizing NADH to NAD⁺. In alcoholic fermentation (e.g. yeast), pyruvate is first decarboxylated to acetaldehyde by pyruvate decarboxylase, then reduced to ethanol by alcohol dehydrogenase, regenerating NAD⁺.